

Components

Components

Arena Design System (ADS)

Version: 1.0

Status: Draft

Overview

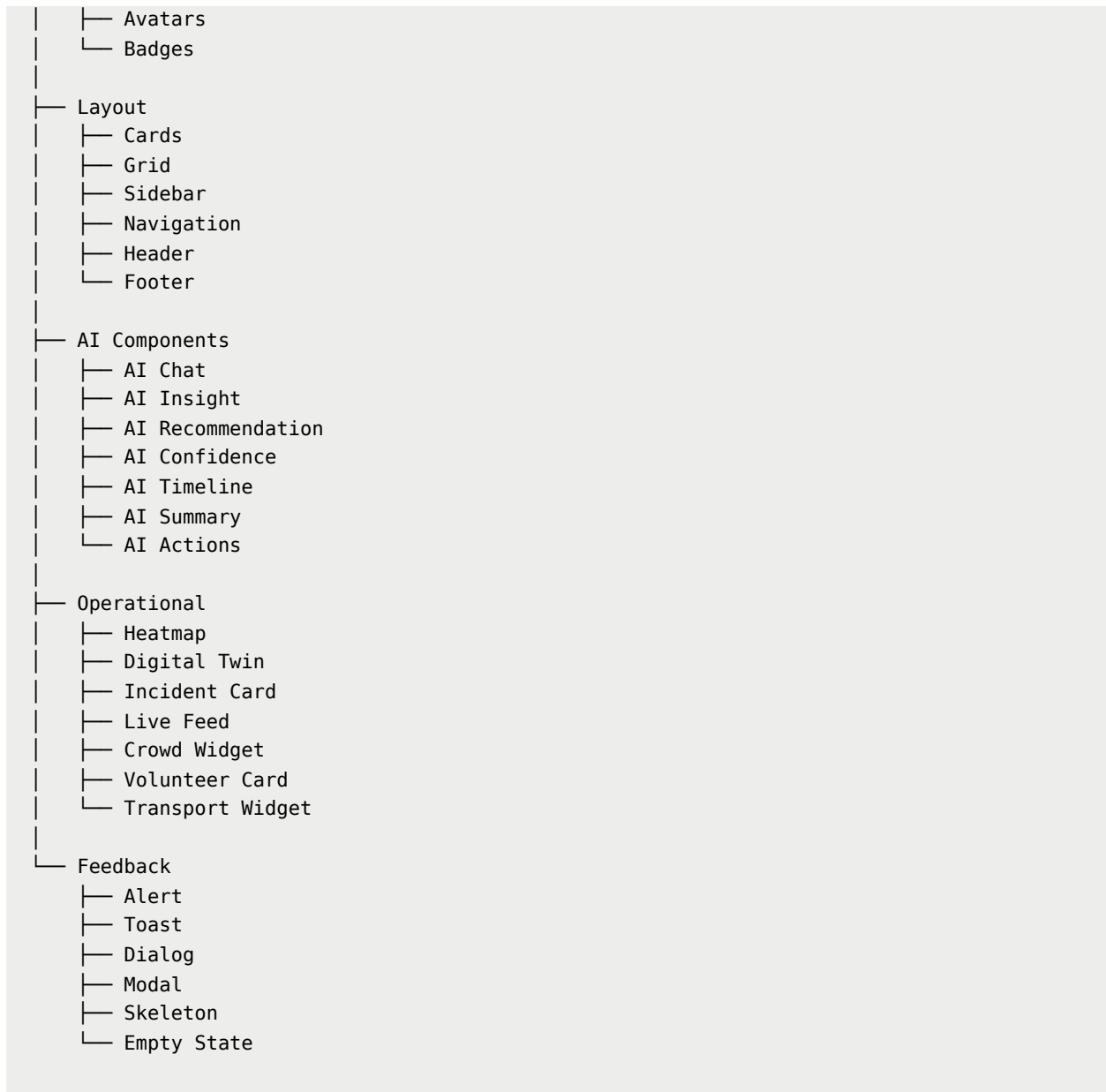
ArenaMind AI is built around reusable, composable UI components.

Every component follows the Arena Design System principles:

- AI-first
- Accessibility-first
- Responsive
- Glassmorphism
- Motion-driven
- Theme-aware
- Reusable
- Scalable

Component Categories

```
Components
├── Foundation
│   ├── Buttons
│   ├── Inputs
│   ├── Typography
│   └── Icons
```



Foundation Components

Button

Buttons initiate user actions.

Variants

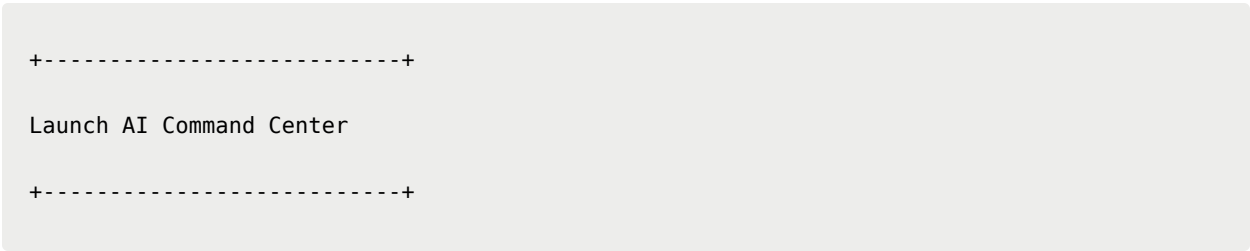
Variant	Usage
Primary	Primary CTA
Secondary	Supporting actions
Ghost	Low emphasis
Danger	Destructive actions
AI	AI interactions
Icon	Compact actions

Primary Button

Purpose

Main action on any screen.

Example



Style

- Gradient background
- Rounded corners
- Soft shadow
- Hover glow
- Animated press

AI Button

Purpose

Triggers AI reasoning.

Examples

Ask AI

Generate Report

Predict Crowd

Explain Incident

Special Effects

- Purple gradient
- Animated border
- Pulsing glow
- Neural particle effect

Icon Button

Used for

- Notifications
- Search
- Settings
- Voice
- Camera

Always circular.

Minimum size:

48 × 48 px

Inputs

Search Input

Search incidents...



Features

- AI autocomplete
- Voice search
- Keyboard shortcuts
- Suggestions

Chat Input

Supports

- Text
- Voice
- Image upload
- Document upload

Example

Ask ArenaMind AI...



Voice Input

Displays

- Live waveform
- Recording animation

- Speech recognition status
-

Cards

Cards are the primary information containers.

Glass Card

Purpose

General container.

Supports

- Charts
- Metrics
- AI
- Maps

Structure

Title

Subtitle

Content

Actions

Metric Card

Purpose

Display KPIs.

Example

Crowd Density

82%

▲ +14%

High

Incident Card

Displays

- Severity
- Timestamp
- Zone
- AI Summary
- Assigned Team
- Status

Example

Security Incident

Gate B

Medium

AI Summary

Unauthorized access detected.

[View Details](#)

AI Insight Card

One of the most important components.

Purpose

Explain AI reasoning.

Structure

Insight
Prediction
Confidence
Evidence
Suggested Action

Example

AI Insight
High congestion predicted.
Confidence
96%
Evidence
• Bus arrival
• Weather
• Ticket scans
Recommendation
Open Gate D.

AI Recommendation Card

Displays

- Recommendation
- Priority
- Impact

- Risk
 - Estimated Benefit
-

AI Confidence Badge

Purpose

Communicate model confidence.

Example

98%

Very High

Color

Green

Above 90%

Yellow

70–89%

Red

Below 70%

Dashboard Components

Stat Widget

Compact KPI.

Visitors

42,900

+6%

Trend Widget

Displays

- Metric
 - Sparkline
 - Delta
-

Live Activity Feed

Shows

- Incidents
- Alerts
- AI actions
- Volunteer updates

Newest always on top.

Notification Center

Grouped by

- Critical
- Warning
- Information
- Success

Supports

- Read
 - Archive
 - Pin
-

AI Components

These components make ArenaMind AI unique.

AI Chat Panel

Purpose

Primary interaction interface.

Supports

- Voice
- Image
- Documents
- Multilingual
- Streaming responses

Layout

Conversation

Suggestions

Actions

Input

AI Summary Panel

Automatically summarizes

- Incidents
- Reports
- Operations
- Crowd status

Structure

Summary

Key Points

Risks

Actions

AI Timeline

Visualizes reasoning.

Event

↓

Detection

↓

Prediction

↓

Recommendation

↓

Resolution

AI Suggested Actions

Displays

Open Gate D

Assign Volunteers

Notify Security

Generate Announcement

Each action is executable.

AI Risk Meter

Visual indicator.

Levels

- Low
 - Medium
 - High
 - Critical
-

AI Explanation Panel

Every prediction should explain

Why?

Evidence?

Confidence?

Alternatives?

Transparency builds trust.

Operational Components

Digital Twin Viewer

Primary visualization.

Features

- Rotate
- Zoom
- Select zone
- Live overlays
- AI predictions

Supports

- Three.js
 - React Three Fiber
-

Crowd Heatmap

Displays

- Density
- Flow
- Prediction
- Congestion

Color Scale

Green

↓

Yellow

↓

Orange

↓

Red

Stadium Map

Layers

- Gates
 - Parking
 - Food
 - Medical
 - Toilets
 - Volunteers
 - Security
 - Crowd
-

Transport Widget

Displays

- Metro
- Parking
- Ride Share
- Shuttle
- Walking

Updates in real time.

Volunteer Card

Displays

- Name
- Location
- Task

- Status
 - ETA
-

Accessibility Widget

Displays

- Accessible routes
 - Elevators
 - Wheelchair seating
 - Audio navigation
-

Sustainability Card

Metrics

- Carbon
- Water
- Energy
- Waste

AI recommendations included.

Charts

Supported

- Line
- Area
- Bar
- Radar
- Heatmap

- Sankey
- Treemap

Animated.

Responsive.

Modals

Used sparingly.

Variants

- Confirmation
 - Incident
 - AI Report
 - Settings
 - Emergency
-

Toasts

Position

Top Right

Duration

4 seconds

Types

Success

Warning

Error

Information

Empty States

Every empty state should educate.

Example

```
No active incidents.  
  
Everything is running smoothly.  
  
Enjoy the match.
```

Loading Components

Never display blank screens.

Supported

- Skeleton
- AI Thinking
- Progress Ring
- Pulse Animation

Component States

Every interactive component supports

- Default
- Hover
- Active
- Focus
- Disabled
- Loading
- Success

- Error
-

Accessibility Requirements

Every component must support

- Keyboard navigation
- Screen readers
- Voice control
- Focus visibility
- High contrast
- Reduced motion

Minimum touch target

48 × 48 px

Responsive Behavior

Desktop

Full component

↓

Tablet

Compact

↓

Mobile

Stacked

↓

Watch

Minimal

Naming Convention

```
ButtonPrimary
ButtonAI
GlassCard
StatCard
IncidentCard
InsightCard
HeatmapWidget
TransportWidget
AIChatPanel
DigitalTwinViewer
VolunteerCard
NotificationItem
ToastSuccess
ModalIncident
SkeletonCard
```

Component Development Standards

Every component must be:

- Reusable
- Typed (TypeScript)
- Accessible (WCAG 2.2 AA)
- Theme-aware
- Dark mode compatible

- Motion-enabled
 - Unit tested
 - Storybook documented
-

Tech Stack Mapping

Component	Technology
UI	React + Next.js
Styling	Tailwind CSS
Variants	Class Variance Authority (CVA)
Icons	Lucide React
Motion	Framer Motion
3D	React Three Fiber + Drei
Charts	Recharts
Maps	Google Maps SDK
AI Chat	Gemini + Groq
State	Zustand
Forms	React Hook Form
Validation	Zod

Future Components

- AI Copilot Sidebar
- Multi-Agent Workspace
- Live Collaboration Cursors
- Predictive Timeline
- Voice Command Dock
- AI Workflow Builder

- Command Palette
 - Scenario Simulator
 - Digital Twin Editor
 - Stadium Replay Viewer
-

Arena Design System

Build once. Reuse everywhere.

Every component should be composable, explainable, and AI-native.